Integration by Substitution

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: u-substitution**

Evaluate each integral using substitution:

a $\int 2x(x^2+1)^5\,dx$ $u = x^2+1$: $= \frac{(x^2+1)^6}{6} + C$.

b $\int \frac{x}{\sqrt{x^2+9}}\,dx$ $u = x^2+9$: $= \sqrt{x^2+9} + C$.

c $\int \cos(3x)\,dx$ $u = 3x$: $= \frac{1}{3}\sin(3x) + C$.

(a) $u = x^2+1$: $= \frac{(x^2+1)^6}{6} + C$.; (b) $u = x^2+9$: $= \sqrt{x^2+9} + C$.; (c) $u = 3x$: $= \frac{1}{3}\sin(3x) + C$.

2 **[Foundational]** • **Target: u-substitution**

Evaluate $\int x^2 e^{x^3}\,dx$.

$u = x^3$, $du = 3x^2\,dx$: $= \frac{1}{3}e^{x^3} + C$.

3 **[Foundational]** • **Target: Definite integrals with substitution**

Evaluate $\int_0^2 x\sqrt{x^2+1}\,dx$ exactly, changing the limits with $u = x^2+1$.

u goes from 1 to 5: $\frac{1}{2}\int_1^5 \sqrt{u}\,du = \frac{1}{2}\left[\frac{2}{3}u^{3/2}\right]_1^5 = \frac{1}{3}(5\sqrt{5} - 1)$.

4 **[Foundational]** • **Target: Definite integrals with substitution**

Evaluate $\int_0^{\pi/2} \sin^3 x \cos x\,dx$.

$u = \sin x$, limits 0 to 1: $\int_0^1 u^3\,du = \frac{1}{4}$.

5 **[Foundational]** • **Target: Definite integrals with substitution**

Evaluate $\int_1^e \frac{\ln x}{x}\,dx$.

$u = \ln x$, limits 0 to 1: $\int_0^1 u\,du = \frac{1}{2}$.

6 **[Foundational]** • **Target: Substitution with trigonometric and exponential integrands**

Evaluate $\int \frac{2x+3}{x^2+3x+1}\,dx$, recognizing the numerator as the derivative of the denominator.

$u = x^2+3x+1$, $du = (2x+3)\,dx$: $= \ln|x^2+3x+1| + C$.

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7 [Foundational] • Target: Choosing the substitution

For each integral, state a suitable substitution u and evaluate:

a $\int x^2(x^3-1)^7\,dx$ $u = x^3-1$, $du = 3x^2dx$: $= \frac{1}{3}\int u^7\,du = \frac{(x^3-1)^8}{24} + C$.

b $\int \tan x\,dx = \int \frac{\sin x}{\cos x}\,dx$ $u = \cos x$, $du = -\sin x\,dx$: $= -\ln|\cos x| + C = \ln|\sec x| + C$.

(a) $u = x^3-1$, $du = 3x^2dx$: $= \frac{1}{3}\int u^7\,du = \frac{(x^3-1)^8}{24} + C$.; (b) $u = \cos x$, $du = -\sin x\,dx$: $= -\ln|\cos x| + C = \ln|\sec x| + C$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Substitution with radicals

Evaluate $\int x\sqrt{2-x}\,dx$ using the substitution $u = 2-x$ (so $x = 2-u$).

$u = 2-x$, $x = 2-u$, $dx = -du$:

$\int (2-u)\sqrt{u}(-du) = \int (u-2)\sqrt{u}\,du = \int (u^{3/2} - 2u^{1/2})du = \frac{2}{5}u^{5/2} - \frac{4}{3}u^{3/2} + C = \frac{2}{5}(2-x)^{5/2} - \frac{4}{3}(2-x)^{3/2} + C$.

9 [Intermediate] • Target: Substitution with radicals

Evaluate $\int_0^3 \frac{x}{\sqrt{x+1}}\,dx$ using $u = x+1$ (so $x = u-1$), changing the limits.

u goes from 1 to 4:

$\int_1^4 \frac{u-1}{\sqrt{u}}\,du = \int_1^4 (u^{1/2} - u^{-1/2})du = \left[\frac{2}{3}u^{3/2} - 2u^{1/2}\right]_1^4 = \left(\frac{16}{3} - 4\right) - \left(\frac{2}{3} - 2\right) = \frac{4}{3} + \frac{4}{3} = \frac{8}{3}$.

10 [Intermediate] • Target: Visualizing an area found by substitution

The graph shows $f(x) = x\sqrt{x^2+1}$ on $[0, 2]$. Evaluate $\int_0^2 x\sqrt{x^2+1}\,dx$ exactly using $u = x^2+1$, and match it to the shaded area.

u goes from 1 to 5: $\frac{1}{2}\int_1^5 \sqrt{u}\,du = \frac{1}{2}\left[\frac{2}{3}u^{3/2}\right]_1^5 = \frac{1}{3}\left(5\sqrt{5} - 1\right) \approx 3.393$, which equals the shaded area under the curve.

11 [Intermediate] • Target: Substitution with double angle and reciprocal forms

Evaluate $\int \sin x\cos x\,dx$ two ways: (a) using $u = \sin x$, and (b) using the identity $\sin x\cos x = \frac{1}{2}\sin(2x)$. Confirm both give equivalent answers.

(a) $u = \sin x$: $\int u\,du = \frac{\sin^2 x}{2} + C$. (b) $\int \frac{1}{2}\sin(2x)\,dx = -\frac{1}{4}\cos(2x) + C$. Using $\cos(2x) = 1 - 2\sin^2 x$: $-\frac{1}{4}(1 - 2\sin^2 x) + C = \frac{\sin^2 x}{2} + C'$ — the same family, differing only by the constant.

12 [Intermediate] • Target: Substitution with double angle and reciprocal forms

Evaluate $\int \frac{1}{x\ln x}\,dx$ using $u = \ln x$.

$u = \ln x$, $du = \frac{1}{x}dx$: $\int \frac{1}{u}\,du = \ln|u| + C = \ln|\ln x| + C$.

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13 [Intermediate] • Target: Substitution in motion and rate problems

A particle's velocity is $v(t) = t\sin(t^2)$ m/s. Find its displacement from $t = 0$ to $t = \sqrt{\pi}$, using $u = t^2$.

$u = t^2$, $du = 2t\,dt$, limits 0 to π: $\frac{1}{2}\int_0^{\pi} \sin u\,du = \frac{1}{2}[-\cos u]_0^{\pi} = \frac{1}{2}(1+1) = 1$ m.

14 [Intermediate] • Target: Substitution in motion and rate problems

Evaluate $\int \frac{(\ln x)^2}{x}\,dx$ using $u = \ln x$.

$u = \ln x$, $du = \frac{1}{x}dx$: $\int u^2\,du = \frac{u^3}{3} + C = \frac{(\ln x)^3}{3} + C$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Recognizing when substitution simplifies a problem • Cross-Lesson: 2.01 Evaluating Limits

Evaluate $\int (x+1)\sqrt{x^2+2x+5}\,dx$ by noticing the numerator is (up to a constant) the derivative of the expression under the root.

$u = x^2 + 2x + 5$, $du = (2x+2)dx = 2(x+1)dx$: $\frac{1}{2}\int \sqrt{u}\,du = \frac{1}{2}\cdot\frac{2}{3}u^{3/2} + C = \frac{1}{3}(x^2+2x+5)^{3/2} + C$.

16 [Challenging] • Target: Substitution with trigonometric and exponential integrands • Cross-Lesson: 2.02 Continuity

Net change in population from $t = 0$ to $t = T$ with rate $r(t)$:.

Final Answer: $\int_0^T r(t)\,dt$. Applying Applications of Integration principles, Net change = integral of rate. Therefore, the verified correct option is $\int_0^T r(t)\,dt$.

17 [Challenging] • Target: Choosing the substitution • Cross-Lesson: 3.01 The Derivative from First Principles

$\int_a^b [f(x) - g(x)]dx$ gives area between curves when:.

Final Answer: $f(x) \geq g(x)$ on $[a, b]$. Applying Applications of Integration principles, Area when $f \geq g$. Therefore, the verified correct option is $f(x) \geq g(x)$ on $[a, b]$.

18 [Challenging] • Target: Substitution with radicals • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

At what rate is the area of a square changing when its side is 4 m and growing at 2 m/s? Show your complete algebraic work.

Final Answer: 16 m²/s. Applying Applications of Derivatives principles, $dA/dt = 2s \cdot ds/dt = 2(4)(2) = 16$. Therefore, the verified correct option is 16 m²/s.

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19 [Challenging] • Target: Visualizing an area found by substitution • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Under the change of variables $x = 2u - v$, $y = 2u + 4v$, a region S in the uv-plane with area 6 maps to a region R. Find the area of R.

Final Answer: 60. $\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = |10|$, so $\text{Area}(R) = |J| \cdot \text{Area}(S) = 60$.

20 [Challenging] • Target: Substitution with double angle and reciprocal forms • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Under the change of variables $x = 2u$, $y = -2u + 4v$, a region S in the uv-plane with area 2 maps to a region R. Find the area of R.

Final Answer: 16. $\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = |8|$, so $\text{Area}(R) = |J| \cdot \text{Area}(S) = 16$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Basic u-Substitution

Evaluate $\int 2x(x^2+5)^4\,dx$.

a. $\frac{1}{5}(x^2+5)^5 + C$ ✓ **b.** $\frac{1}{4}(x^2+5)^4 + C$

c. $(x^2+5)^5 + C$ **d.** $\frac{2}{5}(x^2+5)^5 + C$

Let $u = x^2+5 \Rightarrow du = 2x\,dx$. $\int u^4\,du = \frac{1}{5}u^5 + C = \frac{1}{5}(x^2+5)^5 + C$.

22 [Foundational] • Target: Basic u-Substitution

Evaluate $\int x^2\sqrt{x^3+1}\,dx$.

a. $\frac{2}{9}(x^3+1)^{3/2} + C$ ✓ **b.** $\frac{2}{3}(x^3+1)^{3/2} + C$

c. $\frac{1}{3}(x^3+1)^{3/2} + C$ **d.** $\frac{3}{2}(x^3+1)^{1/2} + C$

Let $u = x^3+1 \Rightarrow du = 3x^2\,dx$. $\frac{1}{3}\int u^{1/2}\,du = \frac{1}{3}\frac{2}{3}u^{3/2} + C = \frac{2}{9}(x^3+1)^{3/2} + C$.

23 [Foundational] • Target: Basic u-Substitution

Evaluate $\int \frac{x}{x^2+9}\,dx$.

a. $\frac{1}{2}\ln(x^2+9) + C$ ✓ **b.** $\ln(x^2+9) + C$

c. $\frac{1}{3}\arctan(x/3) + C$ **d.** $\frac{1}{2(x^2+9)^2} + C$

Let $u = x^2+9 \Rightarrow du = 2x\,dx$. $\frac{1}{2}\int \frac{1}{u}\,du = \frac{1}{2}\ln(x^2+9) + C$.

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24 [Foundational] • Target: Trigonometric and Exponential Substitutions

Evaluate $\int \sin^3(x)\cos(x)\,dx$.

a. $\frac{1}{4}\sin^4(x)+C$ ✓

b. $\frac{1}{3}\sin^3(x)+C$

c. $-\frac{1}{4}\cos^4(x)+C$

d. $\sin^4(x)+C$

Let $u=\sin x \Rightarrow du=\cos x\,dx$. $\int u^3\,du=\frac{1}{4}u^4+C=\frac{1}{4}\sin^4(x)+C$.

25 [Foundational] • Target: Trigonometric and Exponential Substitutions

Evaluate $\int \tan(x)\,dx$.

a. $\ln|\sec x|+C$ ✓

b. $-\ln|\sec x|+C$

c. $\sec^2(x)+C$

d. $\ln|\sin x|+C$

$\int \frac{\sin x}{\cos x}\,dx$. Let $u=\cos x$, $du=-\sin x\,dx$. $-\int \frac{du}{u}=-\ln|\cos x|+C=\ln|\sec x|+C$.

26 [Foundational] • Target: Trigonometric and Exponential Substitutions

Evaluate $\int e^{3x-1}\,dx$.

a. $\frac{1}{3}e^{3x-1}+C$ ✓

b. $3e^{3x-1}+C$

c. $e^{3x-1}+C$

d. $\frac{1}{3}e^{3x}+C$

Let $u=3x-1 \Rightarrow du=3\,dx$. $\frac{1}{3}\int e^u\,du=\frac{1}{3}e^{3x-1}+C$.

27 [Foundational] • Target: Definite Integrals with Limits Change

Evaluate $\int_0^1 x(x^2+1)^3\,dx$.

a. 15/8 ✓

b. $16/8=2$

c. 7/8

d. 15/4

$u=x^2+1$. When $x=0, u=1$; when $x=1, u=2$. $\frac{1}{2}\int_1^2 u^3\,du=\frac{1}{2}\left[\frac{u^4}{4}\right]_1^2=\frac{1}{8}[16-1]=\frac{15}{8}$.

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28 [Foundational] • Target: Definite Integrals with Limits Change

Evaluate $\int_0^{\sqrt{\pi}} x\sin(x^2)\,dx$.

a. 1 ✓

b. 0

c. 2

d. 1/2

$u = x^2$, $du = 2x\,dx$. Bounds: $x = 0 \Rightarrow u = 0$; $x = \sqrt{\pi} \Rightarrow u = \pi$.
$\frac{1}{2}\int_0^{\pi} \sin u\,du = \frac{1}{2}[-\cos u]_0^{\pi} = \frac{1}{2}[-(-1) - (-1)] = \frac{1}{2}(2) = 1$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Advanced Linear Substitutions

Evaluate $\int x\sqrt{x+2}\,dx$.

a. $\frac{2}{5}(x+2)^{5/2} - \frac{4}{3}(x+2)^{3/2} + C$ ✓

b. $\frac{2}{3}(x+2)^{3/2} + C$

c. $\frac{1}{5}(x+2)^{5/2} - \frac{2}{3}(x+2)^{3/2} + C$

d. $\frac{2}{5}x^{5/2} + C$

Let $u = x+2 \Rightarrow x = u-2$ and $dx = du$. $\int (u-2)u^{1/2}\,du = \int (u^{3/2} - 2u^{1/2})\,du = \frac{2}{5}u^{5/2} - \frac{4}{3}u^{3/2} + C$.

30 [Intermediate] • Target: Advanced Linear Substitutions

Evaluate $\int \frac{x}{\sqrt{x+1}}\,dx$.

a. $\frac{2}{3}(x+1)^{3/2} - 2\sqrt{x+1} + C$ ✓

b. $\frac{1}{3}(x+1)^{3/2} - \sqrt{x+1} + C$

c. $2\sqrt{x+1} + C$

d. $\frac{2}{3}(x+1)^{3/2} + C$

Let $u = x+1 \Rightarrow x = u-1$, $dx = du$. $\int (u-1)u^{-1/2}\,du = \int (u^{1/2} - u^{-1/2})\,du = \frac{2}{3}u^{3/2} - 2u^{1/2} + C$.

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31 [Intermediate] • Target: Advanced Linear Substitutions

Evaluate $\int \frac{e^{2x}}{e^x+1}\,dx$.

a. $e^x - \ln(e^x+1) + C$ ✓

b. $e^x + \ln(e^x+1) + C$

c. $\ln(e^x+1) + C$

d. $\frac{1}{2}e^{2x} + C$

Let $u = e^x + 1 \Rightarrow e^x = u - 1$ and $du = e^x\,dx$.
$\int \frac{u-1}{u}\,du = \int (1 - 1/u)\,du = u - \ln|u| + C = e^x + 1 - \ln(e^x+1) + C = e^x - \ln(e^x+1) + C'$.

32 [Intermediate] • Target: Definite Integrals with Limits Change

Under the change of variables $x = 3u + v$, $y = 3u + 3v$, a region S in the uv-plane with area 3 maps to a region R. Find the area of R.

a. 3

b. 6

c. 18 ✓

d. $\frac{1}{2}$

$\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = |6|$, so $\text{Area}(R) = |J| \cdot \text{Area}(S) = 18$.

33 [Intermediate] • Target: Advanced Linear Substitutions

Under the change of variables $x = 4u + v$, $y = 3u + 3v$, a region S in the uv-plane with area 4 maps to a region R. Find the area of R.

a. 36 ✓

b. 4

c. 9

d. 28

$\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = |9|$, so $\text{Area}(R) = |J| \cdot \text{Area}(S) = 36$.

34 [Intermediate] • Target: u-substitution

Under the change of variables $x = 3u + 2v$, $y = u + 2v$, a region S in the uv-plane with area 5 maps to a region R. Find the area of R.

a. 20 ✓

b. 5

c. 4

d. 25

$\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = |4|$, so $\text{Area}(R) = |J| \cdot \text{Area}(S) = 20$.

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35 [Intermediate] • Target: Definite integrals with substitution

Under the change of variables $x = 4u + v$, $y = -u + v$, a region S in the uv-plane with area 6 maps to a region R. Find the area of R.

a. 6

b. 5

c. 30 ✓

d. $\frac{6}{5}$

$\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = |5|$, so $\text{Area}(R) = |J| \cdot \text{Area}(S) = 30$.

36 [Intermediate] • Target: Substitution with trigonometric and exponential integrands

Under the change of variables $x = 3u - 2v$, $y = -u + 2v$, a region S in the uv-plane with area 4 maps to a region R. Find the area of R.

a. 4

b. 20

c. 1

d. 16 ✓

$\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = |4|$, so $\text{Area}(R) = |J| \cdot \text{Area}(S) = 16$.

37 [Intermediate] • Target: Choosing the substitution

Use a double integral to find the area of the region under $y = x^2$ for $0 \le x \le 1$.

a. $\frac{2}{3}$

b. $\frac{1}{3}$ ✓

c. $\frac{1}{6}$

d. $\frac{1}{2}$

$\int_0^1 \int_0^{x^2} 1 \, dy \, dx = \frac{1}{3}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Substitution with radicals • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Weierstrass substitution $t = \tan(x/2)$: $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$. Converts rational trig integrands to:

a. trig of t

b. **rational functions of t** ✓

c. exponentials of t

d. polynomials

Weierstrass: rationalises all trig expressions.

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39 [Challenging] • Target: Visualizing an area found by substitution • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Consider the region R in the xy-plane bounded by the curves $y = x^2$ and $y = 4$. What is the area of this region?

a. $\frac{32}{3}$ ✓

b. $\frac{16}{3}$

c. 8

d. 16

The area of the region bounded by two curves $y = f(x)$ and $y = g(x)$ from $x = a$ to $x = b$, where $f(x) \geq g(x)$ on $[a, b]$, is given by the integral $\int_a^b (f(x) - g(x))dx$. First, we find the intersection points of $y = x^2$ and $y = 4$ by setting them equal: $x^2 = 4$, which gives $x = -2$ and $x = 2$. So, our limits of integration are $a = -2$ and $b = 2$. On the interval $[-2, 2]$, the line $y = 4$ is above the parabola $y = x^2$. Therefore, the area A is given by: $A = \int_{-2}^{2} (4 - x^2)dx$ Now, we eva.

40 [Challenging] • Target: Substitution with double angle and reciprocal forms • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Evaluate $\int_0^2 \int_y^2 e^{x^2}\,dx\,dy$ by reversing the order of integration.

a. $-\frac{1}{2} + \frac{e^4}{2}$ ✓

b. $-1 + e^4$

c. $\frac{1}{2} + \frac{e^4}{2}$

d. $-\frac{1}{4} + \frac{e^4}{4}$

The inner integral has no elementary antiderivative as written. Reversing the order gives $\int_0^2 \int_0^x e^{x^2}\,dy\,dx$, which evaluates to $-\frac{1}{2} + \frac{e^4}{2}$.

41 [Challenging] • Target: Substitution in motion and rate problems • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Evaluate $\int_0^{\sqrt{\pi}} \int_x^{\sqrt{\pi}} \sin(y^2)\,dy\,dx$ by reversing the order of integration.

a. 1 ✓

b. 2

c. $\frac{1}{2}$

d. 3

The inner integral has no elementary antiderivative as written. Reversing the order gives $\int_0^{\sqrt{\pi}} \int_0^y \sin(y^2)\,dx\,dy$, which evaluates to 1.

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42 [Challenging] • Target: Recognizing when substitution simplifies a problem • Cross-Lesson: 6.01 Related Rates

A ladder 13 m long leans against a wall. Base slides at 2 m/s. Rate of top falling when base is 5 m from wall:

a. $-\frac{5}{6}$ m/s ✓

b. $-\frac{5}{12}$ m/s

c. $-\frac{2}{5}$ m/s

d. $-\frac{1}{2}$ m/s

From $x^2 + y^2 = 169$, when $x = 5$ we have $y = 12$. Differentiating gives $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -\frac{5(2)}{12} = -\frac{5}{6}$ m/s.

43 [Challenging] • Target: Basic u-Substitution • Cross-Lesson: 6.02 Optimization and Curve Sketching

Region $x \geq 0$, $y \geq 0$, $x + y \leq 1$: volume when rotated about y-axis by shell method:

a. $\frac{\pi}{3}$ ✓

b. $\frac{\pi}{4}$

c. $\frac{\pi}{2}$

d. $\frac{\pi}{6}$

By cylindrical shells about the y-axis, $V = 2\pi\int_0^1 x(1-x)\,dx = 2\pi\left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = 2\pi\left(\frac{1}{6}\right) = \frac{\pi}{3}$.

44 [Challenging] • Target: Trigonometric and Exponential Substitutions • Cross-Lesson: 7.02 Integration by Parts

Area below x-axis from -2 to 0 of $f(x) = x^2 - 4$:

a. $-\frac{16}{3}$

b. $\frac{8}{3}$

c. $\frac{16}{3}$ ✓

d. 4

Since $x^2 - 4 \leq 0$ on $[-2, 0]$, the area is $\int_{-2}^{0}(4 - x^2)\,dx = \left[4x - \frac{x^3}{3}\right]_{-2}^{0} = 0 - \left(-8 + \frac{8}{3}\right) = \frac{16}{3}$.

45 [Challenging] • Target: Definite Integrals with Limits Change • Cross-Lesson: 8.01 Area Between Curves

Washer method: volume when $R(x)$ is outer and $r(x)$ inner radius:

a. $\pi\int(R - r)^2 dx$

b. $\pi\int(R^2 - r^2)dx$

c. $2\pi\int(R - r)dx$

d. $\pi\int R^2 dx - \pi\int r^2 dx$ **(same)** ✓

Applying Applications of Integration principles, Washer: $\pi\int(R^2 - r^2)dx$. Therefore, the verified correct option is $\pi\int R^2 dx - \pi\int r^2 dx$ (same).

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